

Section 15 Solution

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4 Exercises

4.1 Exercise - Monopoly with Fixed Cost (Problem 1, MT2, Spring 2014)

Consider the case of a monopolist producing quantity Q with total cost $C(Q) = F + cQ$, with $c > 0$ and $F \geq 0$.

1. Comment briefly on the cost function. Does it involve a fixed cost?

Answer:

The cost function involves a constant marginal cost c and a fixed cost F .

2. Determine the marginal cost function C'_q and the average cost function $C(Q)/Q$, and plot the two functions in a graph with x-axis quantity Q and y-axis cost/price.

Answer:

The marginal cost $C'_q = c$, so the marginal cost is constant, and the average cost $C(Q)/Q = F/Q + c$ is decreasing, asymptoting to c for very large Q .

3. Assume now that aggregate demand is given by the linear (inverse) demand function $p(Q) = A - bQ$, with $A > c$. Draw it in the graph with the marginal cost function of point (2). If you want, assume $A = 10$, $c = 5$, $b = 1$, $F = 1$ (just for the graph) Also, derive the marginal revenue MR function, by differentiating with respect to Q the revenue R , and draw the $MR = R'$ function in the graph.

Answer:

The marginal revenue is

$$\frac{\partial}{\partial Q} (Q(A - bQ)) = A - 2bQ,$$

which is a line with the same intercept, but twice as steep.

4. Find graphically the solution for monopoly quantity and price

Answer:

The monopoly solution for quantity is at the intersection of MR and C' curves, then one finds the price on the demand curve.

5. Now solve analytically for the monopoly solution, by maximizing

$$\max_Q Q \cdot p(Q) - C(Q) = Q(A - bQ) - F - cQ.$$

Obtain the first order conditions and solve for Q^* and p^* , the monopoly solution.

Answer:

The solution leads to first order condition

$$A - 2bQ - c = 0,$$

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which is just marginal revenue equal to marginal cost.

The solution is

$$Q_m^* = \frac{A - c}{2b}$$

and the price is

$$p_m^* = A - bQ_m^* = A - b \frac{A - c}{2b} = \frac{A + c}{2}.$$

6. Compute now the profits analytically, plugging in Q^* and p^* in the solution above.

Answer:

The profits are

$$\begin{aligned} Q(A - bQ) - F - cQ &= \frac{A - c}{2b} \left(\frac{A + c}{2} \right) - F - c \frac{A - c}{2b} \\ &= \frac{A - c}{2b} \left(\frac{A - c}{2} \right) - F = \frac{(A - c)^2}{4b} - F \end{aligned}$$

7. A consultant to the government claims, regarding this monopoly: ‘A big problem with monopoly in this industry is that profits are unfairly high.’ Is that always true? Can you give conditions under which monopoly profits in this case would be *zero*?

Answer:

The statement is not always true. Monopoly profits are zero if fixed cost are such that they are equal to the difference between revenue and variable cost, i.e.,

$$F^{\pi=0} = Q_m^*(A - bQ_m^*) - cQ_m^*.$$

In this case, that occurs for

$$F^{\pi=0} = \frac{(A - c)^2}{4b}.$$

Graphically this is where the AC function is tangent to the demand function at Q_m^* .

4.2 Price Discrimination Exercise

Suppose that a monopoly producer of widgets has a constant marginal cost of $MC = 6$ and sells its products in two separated markets whose inverse demand functions are $P_1 = 24 - Q_1$ and $P_2 = 12 - 0.5Q_2$. Notice that consumers in market 1 are more eager to buy than are consumers in market 2 in the sense that the former are willing to pay more for any given quantity.

1. Calculate the monopolist’s profit-maximizing price-quantity combinations in those two markets. What are the total profits earned by the monopolist from these two markets? Compute the deadweight losses in the two markets.

Answer:

The monopolist’s profit-maximizing solution for quantity is at the intersection of MR and MC curves, then one finds the price on the respective demand curve.

$$\begin{array}{ll} MC = 6 & MC = 6 \\ MR_1 = 24 - 2Q_1 & MR_2 = 12 - Q_2 \\ \Rightarrow Q_1^m = 9 & \Rightarrow Q_2^m = 6 \\ \Rightarrow P_1^m = 15 & \Rightarrow P_2^m = 9 \end{array}$$

The profit-maximizing price-quantity combinations in these two markets are: $(P_1^m, Q_1^m) = (15, 9)$ and $(P_2^m, Q_2^m) = (9, 6)$

With this pricing strategy, profits are $\pi = (15 - 6) \cdot 9 + (9 - 6) \cdot 6 = 99$

In order to calculate the deadweight losses we first have to compute the competitive output and price in the two markets. The competitive price is given by $P^c = MC = 6$) in both markets. The competitive outputs are given by substituting in this price in to the respective demand curves. Thus, $Q_1^c = 18$ and $Q_2^c = 12$

We can now compute the deadweight losses:

$$\begin{aligned} DW_1 &= 0.5 \cdot (P_1^m - 6) \cdot (18 - 9) = 40.5 \\ DW_2 &= 0.5 \cdot (P_2^m - 6) \cdot (12 - 6) = 9 \\ \Rightarrow DW &= DW_1 + DW_2 = 49.5 \end{aligned}$$

2. Let's consider now a single-price policy. Calculate the profit maximizing price. How does this pricing policy affect welfare? Compute the total deadweight losses in the two markets. Are both markets served under such a pricing strategy?

Answer:

To solve this problem, we need to aggregate demand to determine the optimal price for the firm.

$$Q_1 = 24 - P \quad \text{and} \quad Q_2 = 24 - 2P$$

So, setting $P \geq 12$ excludes market 2.

If we define aggregate demand as $Q = Q_1 + Q_2$, then aggregate demand will be given by:

$$Q(P) = \begin{cases} 24 - P & \text{if } 24 > P \geq 12 \\ 48 - 3P & \text{if } 12 > P \geq 0 \end{cases}$$

And the inverse demand:

$$P(Q) = \begin{cases} 24 - Q & \text{if } Q < 12 \\ 1/3(48 - Q) & \text{if } Q \geq 12 \end{cases}$$

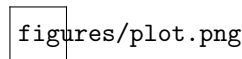
And the revenue function:

$$R(Q) = \begin{cases} 24Q - Q^2 & \text{if } Q < 12 \\ 1/3(48Q - Q^2) & \text{if } Q \geq 12 \end{cases}$$

Let's now calculate marginal revenue:

$$MR(Q) = \begin{cases} 24 - 2Q & \text{if } Q < 12 \\ 1/3(48 - 2Q) & \text{if } Q \geq 12 \end{cases}$$

To calculate firm's optimal quantity we use the optimality condition from the profit maximization process: $MR = MC$. But as we can observe in the figure below MR intersects two times with the MC curve.



How to decide between the two possibilities? We need to calculate which quantity leads to higher profit.

(a) Case $Q < 12$

- $MC = MR \Rightarrow 6 = 24 - 2Q \Rightarrow Q^* = 9 \Rightarrow P^* = 24 - 9 = 15$ from plugging in $Q^* = 9$ into the aggregate demand.
- The profit in this case will be $\pi = (15 - 6)9 = 81$

(b) Case $Q \geq 12$

- $MC = MR \implies 6 = \frac{48}{3} - \frac{2}{3}Q \implies Q^* = 15 \implies P^* = \frac{48}{3} - \frac{1}{3}15 = 11$ from plugging in $Q^* = 15$ into the aggregate demand.
- The profit in this case will be $\pi = (11 - 6)15 = 75$

Thus, the firm realizes a greater profit producing $Q^* = 9$, which leads to $P^* = 15$. Notice that this price excludes market 2 from consumption.

Under a single-price policy the monopoly would simply cease serving market 2, since it can maximize profits by charging a price of 15 and at that price no widgets will be bought in market 2 (because the maximum willingness to pay is 12). Total deadweight loss in this situation is therefore increased from its previous level (49.5) as total potential consumer surplus in market 2 is now lost:

$$\begin{aligned} DW_1 &= 0.5 \cdot (P_1^m - 6) \cdot (18 - 9) = 40.5 \\ DW_2 &= 0.5 \cdot (12 - 6) \cdot (12 - 6) = 18 \\ \Rightarrow DW &= DW_1 + DW_2 = 76.5 \end{aligned}$$

In this case, constraining the monopoly to charge a single price would reduce welfare. This illustrates a situation where third-degree price discrimination is welfare improving over a single price policy as the discriminatory pricing policy permits “smaller markets to be served. Whether such a situation is common is an important policy question. Consider, for example, the case of the U.S. pharmaceutical manufacturers charging higher prices at home than abroad.